CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 1 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. Which of the following solutions will form an ideal solution? (a) Chloroform and acetone (b) Ethanol and acetone (c) n-hexane and n-heptane (d) Phenol and aniline [Solutions] [PYQ Term-1 2021] [Confidence: Very High]

Q2. The relationship between the crystal field splitting energy for octahedral (Δ_o) and tetrahedral (Δ_t) complexes is: (a) $\Delta_t = \frac{4}{9}\Delta_o$ (b) $\Delta_t = \frac{1}{2}\Delta_o$ (c) $\Delta_o = 2\Delta_t$ (d) $\Delta_o = \frac{4}{9}\Delta_t$ [Coordination Compounds] [PYQ 2020] [Confidence: Very High]

Q3. The slope in the plot of $\ln[R]$ vs. time for a first-order reaction is: (a) $\frac{+k}{2.303}$ (b) $-k$ (c) $\frac{-k}{2.303}$ (d) $+k$ [Chemical Kinetics] [PYQ 2020] [Confidence: Very High]

Q4. The intense purple colour of $KMnO_4$ is due to: (a) $d-d$ transitions (b) Charge transfer from ligand to metal (c) Unpaired electrons in the d orbital of Mn (d) Charge transfer from metal to ligand [The d- and f-Block Elements] [PYQ SQP 2022] [Confidence: Very High]

Q5. The oxidation number of Ni in $[Ni(CO)_4]$ is: (a) 0 (b) 2 (c) 3 (d) 4 [Coordination Compounds] [PYQ 2020] [Confidence: Very High]

Q6. Which of the following alkyl halides will undergo an S_N1 reaction most readily? (a) $(CH_3)_3C-Cl$ (b) $(CH_3)_3C-Br$ (c) $(CH_3)_3C-F$ (d) $(CH_3)_3C-I$ [Haloalkanes and Haloarenes] [PYQ SQP Term-1 2021] [Confidence: Very High]

Q7. The major product of the acid-catalysed dehydration of 1-methylcyclohexanol is: (a) 1-methylcyclohexane (b) 1-methylcyclohexene (c) 1-cyclohexylmethanol (d) 1-methylenecyclohexane [Alcohols, Phenols and Ethers] [PYQ SQP 2022] [Confidence: Very High]

Q8. Which of the following compounds will NOT undergo the aldol condensation reaction? (a) CH_3CHO (b) CH_3CH_2CHO (c) CH_3COCH_3 (d) $HCHO$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021] [Confidence: Very High]

Q9. How much charge in terms of Faraday is required for the reduction of 1 mol of Zn^{2+} to Zn? (a) $1\ F$ (b) $2\ F$ (c) $3\ F$ (d) $0.5\ F$ [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q10. Enantiomers (optical isomers) differ from each other only in their: (a) Boiling point (b) Rotation of plane-polarised light (c) Melting point (d) Solubility [Haloalkanes and Haloarenes] [PYQ Term-1 2021] [Confidence: High]

Q11. The correct order of decreasing acidic strength is: (a) $CF_3COOH > CHCl_2COOH > HCOOH > CH_3COOH$ (b) $CH_3COOH > HCOOH > CF_3COOH > CHCl_2COOH$ (c) $HCOOH > CF_3COOH > CHCl_2COOH > CH_3COOH$ (d) $CF_3COOH > CH_3COOH > HCOOH > CHCl_2COOH$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021] [Confidence: Very High]

Q12. What is the secondary valency of Cobalt in $[Co(en)_2Cl_2]^+$? (a) 6 (b) 4 (c) 2 (d) 8 [Coordination Compounds] [PYQ 2023] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): An azeotropic mixture of two liquids has a boiling point lower than either of them when it shows a negative deviation from Raoult's law. Reason (R): In a negative deviation, the intermolecular forces between the components are stronger than in the pure liquids. [Solutions] [PYQ Term-1 2021] [Confidence: High]

Q14. Assertion (A): Chlorobenzene is less reactive towards nucleophilic substitution reactions than haloalkanes. Reason (R): The $C-Cl$ bond in chlorobenzene acquires a partial double bond character due to resonance. [Haloalkanes and Haloarenes] [PYQ Term-1 2021] [Confidence: Very High]

Q15. Assertion (A): $[Pt(en)_2Cl_2]^{2+}$ complex is more stable than $[Pt(NH_3)_4Cl_2]^{2+}$ complex. Reason (R): $[Pt(en)_2Cl_2]^{2+}$ complex shows the chelate effect. [Coordination Compounds] [PYQ 2020] [Confidence: High]

Q16. Assertion (A): Carboxylic acids are more acidic than phenols. Reason (R): Phenols are ortho and para directing. [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2020] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. A first-order reaction takes 20 minutes for 50% completion. Calculate the value of the rate constant (k). [Chemical Kinetics] [PYQ Term-2 2022] [Confidence: Very High]

Q18. Give two points of difference between a double salt and a coordination complex. [Coordination Compounds] [PYQ 2019] [Confidence: High]

Q19. Account for the following: (a) Zinc (Zn, $Z = 30$) is not considered a transition element. (b) The enthalpies of atomisation of transition metals are high. [The d- and f-Block Elements] [Delhi Gov. SQP Term-2 2022] [Confidence: Very High]

Q20. Out of chlorobenzene and benzyl chloride, which one gets more easily hydrolysed by aqueous $NaOH$ to form an alcohol, and why? [Haloalkanes and Haloarenes] [PYQ 2018] [Confidence: Very High]

Q21. Give structural reasons for the following observation: Ortho-nitrophenol is steam volatile, while para-nitrophenol is not steam volatile. [Alcohols, Phenols and Ethers] [PYQ 2023] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. Calculate the mass of ascorbic acid ($C_6H_8O_6$, Molar mass = $176\ g\ mol^{-1}$) to be dissolved in $75\ g$ of acetic acid to lower its freezing point by $1.5°C$. ($K_f = 3.9\ K\ kg\ mol^{-1}$). [Solutions] [PYQ 2020] [Confidence: Very High]

Q23. The limiting molar conductivities (Λ_m°) for $NaCl$, HCl, and CH_3COONa are 126.4, 425.9, and $91.0\ S\ cm^2\ mol^{-1}$ respectively. Calculate the limiting molar conductivity of acetic acid (CH_3COOH). [Electrochemistry] [PYQ 2022] [Confidence: Very High]

Q24. The rate of a reaction becomes four times when the temperature changes from $293\ K$ to $313\ K$. Calculate the energy of activation (E_a) of the reaction assuming that it does not change with temperature. ($R = 8.314\ J\ K^{-1}\ mol^{-1}$, $\log 4 = 0.6021$). [Chemical Kinetics] [PYQ 2019] [Confidence: Very High]

Q25. Explain the following observations: (a) Mn^{2+} is much more resistant than Fe^{2+} towards oxidation. (b) Sc^{3+} is colourless in aqueous solution whereas Ti^{3+} is coloured. (c) Transition metals generally form alloys easily. [The d- and f-Block Elements] [PYQ 2023 / 2020] [Confidence: Very High]

Q26. (a) Arrange the following compounds in the increasing order of their reactivity towards an S_N2 reaction: 2-Bromopentane, 1-Bromopentane, 2-Bromo-2-methylbutane. (b) Why is the thionyl chloride ($SOCl_2$) method preferred for preparing alkyl chlorides from alcohols? [Haloalkanes and Haloarenes] [PYQ 2023 / 2019] [Confidence: High]

Q27. Write the detailed mechanism (using curved arrow notation) of the acid-catalysed dehydration of ethanol to yield ethene at $443\ K$. [Alcohols, Phenols and Ethers] [PYQ 2018] [Confidence: Very High]

Q28. Give simple chemical tests to distinctly distinguish between the following pairs of compounds: (a) Propanal and Propanone (b) Benzaldehyde and Benzoic acid (c) Phenol and Ethanol [Aldehydes, Ketones and Carboxylic Acids / Alcohols] [PYQ 2014 / 2017] [Confidence: Very High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Crystal Field Theory Transition metal complexes show distinct colours and magnetic properties depending on the nature of ligands attached to the central metal ion. The crystal field splitting energy (Δ_o) dictates whether a complex will be high-spin or low-spin. Strong field ligands cause a large splitting ($\Delta_o > P$), whereas weak field ligands cause a small splitting ($\Delta_o < P$). (a) Write the electronic configuration of a d^5 ion in an octahedral field if it is attached to a strong field ligand ($\Delta_o > P$). (1 Mark) (b) Which of the two complexes, $[NiCl_4]^{2-}$ or $[Ni(CN)_4]^{2-}$, is diamagnetic and why? (2 Marks) (c) Why are low-spin tetrahedral complexes rarely observed? (1 Mark) [Coordination Compounds] [CBSE Term-2 2022 / Practice] [Confidence: Very High]

Q30. Case Study: Nucleophilic Addition in Carbonyl Compounds Aldehydes and ketones undergo nucleophilic addition reactions. The reactivity of the carbonyl group is influenced by both steric and electronic (inductive) effects. Aldehydes are generally more reactive than ketones. Furthermore, the presence of α-hydrogens allows them to undergo condensation reactions like the Aldol condensation, whereas those lacking α-hydrogens undergo the Cannizzaro reaction when treated with concentrated alkali. (a) Arrange the following in increasing order of reactivity towards nucleophilic addition: Ethanal, Propanone, Formaldehyde, Butanone. (1 Mark) (b) Write the structure of the product formed when Acetaldehyde reacts with dilute $NaOH$ (Aldol condensation). (1 Mark) (c) A mixture of benzaldehyde and formaldehyde is heated with concentrated aqueous $NaOH$. Name the products formed and identify which one undergoes oxidation. (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021 / Practice] [Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) Write the Nernst equation and calculate the e.m.f. of the following cell at $298\ K$: $Mg(s)|Mg^{2+}(0.001\ M)||Cu^{2+}(0.01\ M)|Cu(s)$ Given: $E^\circ_{Mg^{2+}/Mg} = -2.37\ V$, $E^\circ_{Cu^{2+}/Cu} = +0.34\ V$. (3 Marks) (b) Define a fuel cell. Write any two advantages of fuel cells over ordinary batteries. (2 Marks) [Electrochemistry] [PYQ 2019 / 2018] [Confidence: Very High]

Q32. (a) Describe the preparation of Potassium permanganate ($KMnO_4$) from pyrolusite ore (MnO_2). Write the balanced chemical equations for the two major steps involved. (3 Marks) (b) Using Valence Bond Theory (VBT), deduce the state of hybridisation, geometry, and magnetic behaviour of the complex $[Co(NH_3)_6]^{3+}$. (Atomic no. of Co = 27). (2 Marks) [The d- and f-Block Elements / Coordination Compounds] [PYQ 2023 / 2014] [Confidence: High]

Q33. An organic compound **A** with molecular formula C_8H_8O forms an orange-red precipitate with 2,4-DNP reagent and gives a yellow precipitate of compound **B** when heated with iodine in the presence of sodium hydroxide. It neither reduces Tollen's reagent nor Fehling's reagent. On drastic oxidation with chromic acid, compound **A** gives a carboxylic acid **C** having the molecular formula $C_7H_6O_2$. (a) Identify the exact structures and IUPAC names of compounds **A**, **B**, and **C**. (3 Marks) (b) Write the chemical equation for the reaction of compound **A** with iodine and $NaOH$ (Iodoform reaction). (1 Mark) (c) How will you convert compound **C** back to benzene? (1 Mark) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016 / Practice] [Confidence: Very High]

Answer Key

Q1. (c) n-hexane and n-heptane. (Similar intermolecular interactions). **Q2.** (a) $\Delta_t = \frac{4}{9}\Delta_o$. **Q3.** (b) $-k$. **Q4.** (b) Charge transfer from ligand (oxygen) to metal (manganese). **Q5.** (a) 0. (CO is a neutral ligand). **Q6.** (d) $(CH_3)_3C - I$. (Forms a stable tertiary carbocation and iodide is the best leaving group). **Q7.** (b) 1-methylcyclohexene. (Follows Saytzeff's rule to give the more substituted, stable alkene). **Q8.** (d) $HCHO$. (Formaldehyde lacks an α-hydrogen). **Q9.** (b) $2\ F$. (Reaction: $Zn^{2+} + 2e^- \to Zn$). **Q10.** (b) Rotation of plane-polarised light. **Q11.** (a) $CF_3COOH > CHCl_2COOH > HCOOH > CH_3COOH$. (Electron-withdrawing halogens increase acidity via -I effect). **Q12.** (a) 6. (Ethylenediamine 'en' is a bidentate ligand: $2 \times 2 = 4$, plus two chlorides = 6). **Q13.** (d) A is false but R is true. (A negative deviation forms a *maximum* boiling azeotrope, not lower). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. (Bidentate 'en' ligands form stable chelate rings). **Q16.** (b) Both A and R are true but R is not the correct explanation of A. (Reason for higher acidity is the resonance stabilization of the carboxylate ion over the phenoxide ion).

Q17. For a first-order reaction, $k = \frac{0.693}{t_{1/2}}$. Here $t_{1/2} = 20$ minutes. $k = \frac{0.693}{20} = 0.03465\ min^{-1}$. **Q18.** 1. Double salts completely dissociate into simple ions in water (e.g., Mohr's salt). Coordination complexes retain their complex identity in solution and do not fully dissociate. 2. In double salts, metal exhibits normal valency, whereas in complexes, it exhibits both primary and secondary valencies. **Q19.** (a) Zn ($3d^{10}4s^2$) has completely filled d-orbitals in its ground state and common oxidation state (+2). (b) Due to the presence of a large number of unpaired d-electrons, transition metals have strong interatomic metallic bonding. **Q20.** Benzyl chloride. It easily hydrolyses via the S_N1 mechanism because the resulting benzyl carbocation is highly stabilized by resonance. Chlorobenzene resists hydrolysis due to the partial double bond character of the C-Cl bond. **Q21.** o-Nitrophenol exhibits intramolecular hydrogen bonding, forming discrete volatile molecules. p-Nitrophenol exhibits intermolecular hydrogen bonding, causing molecular association which increases the boiling point (not steam volatile).

Q22. $\Delta T_f = K_f \times m \implies \Delta T_f = K_f \times \frac{W_B \times 1000}{M_B \times W_A}$. $1.5 = 3.9 \times \frac{W_B \times 1000}{176 \times 75}$. $W_B = \frac{1.5 \times 176 \times 75}{3.9 \times 1000} = 5.076\ g$. **Q23.** According to Kohlrausch's law: $\Lambda^\circ_m(CH_3COOH) = \Lambda^\circ_m(CH_3COONa) + \Lambda^\circ_m(HCl) - \Lambda^\circ_m(NaCl) = 91.0 + 425.9 - 126.4 = 390.5\ S\ cm^2\ mol^{-1}$. **Q24.** $\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$. $0.6021 = \frac{E_a}{19.147} \times \left[\frac{313-293}{293 \times 313}\right] \implies 0.6021 = \frac{E_a \times 20}{19.147 \times 91709}$.

$E_a = \frac{0.6021 \times 19.147 \times 91709}{20} \approx 52,863\ J\ mol^{-1}$ or $52.86\ kJ\ mol^{-1}$. **Q25.** (a) Mn^{2+} has a highly stable exactly half-filled $3d^5$ configuration, resisting further loss of electrons. Fe^{2+} ($3d^6$) readily loses an electron to gain the stable $3d^5$ state. (b) Sc^{3+} has a $3d^0$ configuration (no unpaired electrons for d-d transition). Ti^{3+} is $3d^1$, allowing a d-d transition. (c) They have similar atomic sizes (differing by <15%), allowing atoms of one metal to easily substitute for another in the crystal lattice. **Q26.** (a) 2-Bromo-2-methylbutane < 2-Bromopentane < 1-Bromopentane. (Reactivity for S_N2 decreases with increasing steric hindrance: $3^\circ < 2^\circ < 1^\circ$). (b) The by-products of the $SOCl_2$ reaction (SO_2 and HCl) are gases that easily escape, leaving behind pure alkyl chloride. **Q27.** Step 1: Protonation of ethanol: $CH_3CH_2OH + H^+ \rightleftharpoons CH_3CH_2OH_2^+$. Step 2: Formation of carbocation (slow, rate-determining): $CH_3CH_2OH_2^+ \rightarrow CH_3CH_2^+ + H_2O$. Step 3: Elimination of a proton to form ethene: $CH_3CH_2^+ \rightarrow CH_2 = CH_2 + H^+$. **Q28.** (a) Add Tollens' reagent: Propanal gives a silver mirror, Propanone does not. (b) Add $NaHCO_3$: Benzoic acid gives brisk effervescence of CO_2, Benzaldehyde does not. (c) Add neutral $FeCl_3$: Phenol gives a violet colouration, Ethanol does not.

Q29. (a) $t_{2g}^5 e_g^0$. (b) $[Ni(CN)_4]^{2-}$. CN^- is a strong field ligand, causing the $3d^8$ electrons of Ni^{2+} to pair up, leaving no unpaired electrons (diamagnetic). Cl^- is weak field, leaving 2 unpaired electrons (paramagnetic). (c) Tetrahedral crystal field splitting energy (Δ_t) is naturally smaller ($\approx \frac{4}{9}\Delta_o$). It is rarely larger than the pairing energy (P), so electrons prefer to stay unpaired rather than pair up. **Q30.** (a) Butanone < Propanone < Ethanal < Formaldehyde. (Steric hindrance and +I effect decrease reactivity). (b) $2CH_3CHO \xrightarrow{dil.\ NaOH} CH_3 - CH(OH) - CH_2 - CHO$ (3-Hydroxybutanal / Aldol). (c) Products: Benzyl alcohol ($C_6H_5CH_2OH$) and Sodium formate ($HCOONa$). Formaldehyde is more reactive and undergoes oxidation to formate, while benzaldehyde is reduced to benzyl alcohol (Crossed Cannizzaro).

Q31. (a) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.34 - (-2.37) = 2.71\ V$. $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[Mg^{2+}]}{[Cu^{2+}]}$. Here $n = 2$. $E_{cell} = 2.71 - \frac{0.0591}{2} \log \frac{10^{-3}}{10^{-2}} = 2.71 - 0.0295 \log(10^{-1}) = 2.71 + 0.0295 = 2.7395\ V$. (b) Galvanic cells that convert the energy of combustion of fuels directly into electrical energy. Advantages: Highly efficient (70-75%) and pollution-free.

Q32. (a) Step 1: Fusion with KOH/air to form potassium manganate: $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta} 2K_2MnO_4 + 2H_2O$. Step 2: Electrolytic or acidic oxidation to permanganate: $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$. (b) Co is $3d^7 4s^2$. Co^{3+} is $3d^6$. NH_3 is a strong field ligand for Co^{3+}, causing electron pairing. This leaves two empty 3d orbitals. Hybridisation is d^2sp^3. Geometry is Octahedral. Magnetic behaviour is Diamagnetic (no unpaired electrons). **Q33.** (a) Compound **A**: Acetophenone / 1-Phenylethanone ($C_6H_5COCH_3$). Compound **B**: Iodoform (CHI_3). Compound **C**: Benzoic acid (C_6H_5COOH). (b) $C_6H_5COCH_3 + 3I_2 + 4NaOH \rightarrow C_6H_5COONa + CHI_3 \downarrow + 3NaI + 3H_2O$. (c) Heat Benzoic acid with Soda lime ($NaOH$ and CaO): $C_6H_5COOH + NaOH \xrightarrow{CaO,\ \Delta} C_6H_6 + Na_2CO_3$.
